Exercise 2

Aggregate Functions + Grouping

1) Query to find total number of employees

```
SELECT    COUNT (id) AS total number of employees
FROM      employees - db
```

total number of employees
10

2) ~~SELECT~~ Total Salary paid to all employees in IT

```
SELECT    (SUM) (salary) AS total salary paid in IT
FROM      employees - db
WHERE     department = IT
```

total salary in IT
220 000

3) Average salary of employees in HR

```
SELECT    AVG (salary) AS Average salary of employees in HR
FROM      employees - db
WHERE     department = IT
```

Average salary of employees in HR
49 500

4) Highest and lowest salary in the company

```
SELECT   MAX(salary) AS highest salary
         MIN(salary) AS lowest salary,
FROM     employees_db
```

employees_db

highest salary	lowest salary
62000	48 000

5) SQL query to group employees by department & display the total salary in each department

```
SELECT   department
         SUM(salary) as total_salary
FROM     employees_db
GROUP BY department
```

employees_db

department	total_salary
IT	220 000
HR	98 000
Finance	119 000
Marking	105 000

6) SQL to count how many employees work in each city

```
SELECT   city
         COUNT(id) AS Number_of_employees
FROM     employees_db
```

city	Number_of_employees
New York	2
Chicago	3
Los Angles	2
San Fransico	2
Houston	1

7) Group employees by department, calculate avg salary in each department & order by DESC salary.

```
SELECT      department,
            AVG(salary) AS Average_salary_in_Department
FROM        employees_db
GROUP BY    salary DES
```

employees_db

Department	Average_salary_in_Department
IT	55 000
HR	49 500
Finance	57 500
Marketing	52 500

8 Write a SQL : to find department where the total salary paid exceeds 100 000

```
SELECT      department
            SUM(salary) AS total_salary
FROM        employees_db
WHERE       total_salary > 100 000
```

Department	total salary
IT	220 000
Finance	117 000
Marketing	105 000

9) SELECT city
COUNT ~~SUM~~ (Id) AS N° of employees
FROM employees_db
WHERE N° of employees > 1

employees_db

city	№ of employees
New York	2
Chicago	3
Los Angles	2
San Fransico	2

10 Write a SQL to find the department with the highest average salary

```
SELECT department
       SUM(salary) AS total_average_salary
FROM   employees_db
GROUP BY total_average_salary ASEC
LIMIT  1
```

department	total_average_salary
Finance	59,000